Assiut University

BIS401

An *Awesome* Introduction to Web Development

Lecture 1: Introduction

After this class, you are expected to understand:

- The Internet
- HTML and XHTML
- Tools for the Web Designer
- HTML Skeleton
- Embedded Tags (text, hyperlinks, images, tables, select, input, forms, …)
- Client-Side VS. Server-Side Scripting
- CSS
- Why Keeping Style Separate from Structure and Semantics

We will discover :

- **The Structure of a Page**
- **Understanding the Basic Document Structure**
- **Attributes and Attribute Groups**
- **Basic Text Formatting**
- **Presentational Elements**
- **Lists**
- **Links and Navigation**
- **Colors, Images, and Objects**
- **Tables**

Also We will discover:

- **Introducing Forms**
- **Creating a Form with the <form> Element**
- **Frames**
- **When to Use Frames**
- **Introducing the Frameset**
- **The <frameset> Element**
- **Cascading Style Sheets**
- **Introducing CSS**
- **A Basic Example**
- **Inheritance**
- **Where You Can Add CSS Rules**

Finally:

- Learning JavaScript
- What Is Programming About?
- How to Add a Script to Your Pages
- Starting to Program with JavaScript
- Variables , Operators, Control, Looping
- Form validation.
- VBScripts.
- JavaScript Vs VBScripts
- Server side Programming, JSP, ASP

The Internet

- A worldwide collection of computer networks sharing digital information via a common set of networking and software protocols.

- It began as a military experiment but soon the Internet opened up for business.

- At about the same time, was born Hypertext Markup Language (HTML), browser software, and the Web.

Hyper Text Markup Language (HTML)

- HTML is used to describe the structure of a Webpage.
- HTML and XHTML define the syntax and placement of special, embedded directions that aren't displayed by the browser but advise it how to display the contents of the document, including text, images, and other support media.
- Some parts will show up as a paragraph, others as tables, others as headings, and so on.
- Note that for styling (that describes whether the contents should show up left aligned or centered, bold or italic, red or green, etc.) we will use CSS
- XHTML, provide many different ways to let you define the appearance of your documents, but their focus is on structure, not appearance. ⇒ Of course, appearance is important, And that is why the companion CSS standard is important.

Developing Tools

- Any text editor such as notepad or wordpad or any other editor.
  - The only requirement is that the file extension must be .htm or .html.
- GUI editors
  - Many GUI editors are available that enables you to create HTML pages without writing any codes such as MSFrontpage, Netscape composer, Macromedia Dreamweaver

Why learn HTML ?

- Using a graphical interface to write code for you means you do not learn how to codeWeb pages yourself; the tool generates the code for you and you are not encouraged to study or understand this code.
- Without understanding the code you will not be able to make fine adjustments to the way a page looks.
- Most experienced Web page authors who use WYSIWYG tools still change code by hand before they are satisfied with the appearance of a Web page.

- Most authoring tools write code that deviates from the standards. They will often change capitalization, or order of elements and attributes, and leave out values for attributes.

- Once you have really learned to code in XHTML it is very easy to use any WYSIWYG editor to create Web pages. But it is not easy to write code if you just learn to use a WYSIWYG editor because you are not encouraged to understand the code behind it and your skills are locked into that tool.

Front End Issues

A Web page acts as an interface for your Information. A web page should

1- Use attractive Images and graphical elements.

2-Include constantly updated hyperlinks.

3-Use Tables wisely.

4-use carefully designed forms.

5-Use Images sparingly

6- Easy navigable.

Backend Issues

- Bandwidth and download time:

HTML pages should be kept small in size on the hard disk of the server in order not to take too much time on downloads. Images usually accompany HTML pages and they usually take large size, JPEG and GIF format support image compression.

Our Productions

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نظام التسجيل في دورات التربية العسكرية للذكور والتربية الوطنية للإناث
للعام الدراسي 2024-2025
والعام الدراسي 2025-2026

التربية العسكرية للذكور

تسجيل الطلاب الذكور

تعليمات حضور الدورة

التربية الوطنية للإناث

تسجيل الطالبات الإناث

تعليمات حضور الدورة

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مرحباً بك في منصة خدمات الطلاب

بوابتك الذكية للوصول إلى بياناتك وخدماتك الجامعية بسهولة وسرعة وأمان.

الكشف الطبى للمستجدين

تحصيل نظير الكشف الطبى لمستجدين عام 2026_2025

احجز الان

البيانات الجامعية

استعرض بياناتك الجامعية الرسمية بسهولة.

عرض البيانات

نتيجتي

تُعرض نتائج فقط الكليات التي أفادت بإتاحة النتائج للطلاب عبرنا.

عرض النتائج

طلب الكارنيه الجامعي

قدّم طلب استخراج الكارنيه بسهولة وسرعة.

معاينة

طلب الكارت الذكي

احصل على كارتك الذكي للاستخدام الجامعي.

تحت الإنشاء

بدل تالف للمدينة

طلب استخراج كارنيه بدل تالف للمدينة الجامعية.

تحت الإنشاء

تحميل النماذج الرسمية

قم بتحميل النماذج الإدارية المعتمدة.

تحميل النماذج

الدعم الفني

تواصل مع الدعم لحل أي مشكلة تواجهك.

تواصل معنا

✨ برنامج الترجمة الإنجليزية بنظام الساعات المعتمدة (English Translation Special Program) ✨

اكسر حواجز اللغة وافتح آفاقًا جديدة نحو عالم من الفرص

إعلان القبول

تعلن كلية الآداب - جامعة أسيوط عن بدء قبول دفعة جديدة ببرنامج الترجمة الإنجليزية بنظام الساعات المعتمدة (English Translation Special Program) للعام الجامعي ٢٠٢٥/٢٠٢٦.

مميزات البرنامج

- تدريس الترجمة التحريرية والشفهية في مجالات متنوعة
- تنمية مهارات العمل الجماعي وإعداد العروض التقديمية
- ورش تدريبية متخصصة وفرص تدريب ميداني
- تعليم على أيدي نخبة من أساتذة جامعة أسيوط

مجالات العمل

تأهيل الخريجين للعمل في الجامعات، المدارس الدولية، الشركات العالمية، البنوك، قطاع السياحة، والمؤسسات الإعلامية.

المصروفات

١٥٥٧٤ جنيه سنويًا، مع إمكانية التقسيط على قسطين.

شروط القبول

- شهادة الثانوية العامة أو ما يعادلها
- استثناء شرط درجة اللغة الإنجليزية لهذا العام

منصة التقديم على دبلومات كلية الحاسبات والمعلومات

جامعة أسيوط - العام الجامعي 2025 / 2026

يسعدنا استقبال طلبات التقديم للدبلومات التخصصية عبر منصتنا الإلكترونية التي تمنحك فرصًا واسعة للعمل في مجالات متعددة داخل مصر وخارجها.

ابدأ التقديم الآن | متابعة الطلب

تنويه هام لجميع المتقدمين

يجب على جميع المتقدمين رفع كافة المستندات والمرفقات المطلوبة بشكل كامل، حيث انه لن يتم النظر في أي طلب غير مكتمل. كما نود التوضيح أن **الدبلومة متاحة لخريجي جميع الكليات** وليست مقصورة فقط على خريجي كلية الحاسبات والمعلومات.

تنوع التخصصات

- شبكات الحاسب
- تطوير مواقع الإنترنت
- التجارة الإلكترونية
- أمن شبكات الحاسب

أهمية الدبلومة

تفتح مجالات عمل واسعة في قطاعات المعلومات، البرمجيات، الحكومة الإلكترونية، والتجارة الإلكترونية. كما تطور مهارات تقنية عالية مطلوبة في سوق العمل المحلي والإقليمي.

نظام الدراسة

- عدد الساعات المعتمدة: 36 ساعة.
- محاضرات نظرية وتدريب عملي باستخدام أحدث تقنيات التعليم الإلكتروني.
- مرونة في المواعيد تناسب الموظفين والعاملين.

شروط القبول

يشترط الحصول على بكالوريوس أو ليسانس من إحدى الجامعات المصرية أو ما يعادلها. بالنسبة للتخصصات غير المناظرة، يشترط موافقة مجلس الكلية.

فرص العمل بعد الانتهاء من الدبلومه

العمل الحر (Freelancing)

العمل بشكل مستقل عن بُعد مع شركات محلية وعالمية عبر منصات مثل Upwork و Freelancer أو تقديم خدمات مباشرة في مجالات تطوير الويب والتطبيقات.

قطاع البنوك

العمل في أقسام تقنية المعلومات وإدارة قواعد البيانات وتطوير الأنظمة البنكية الإلكترونية.

قطاع البرمجيات

تطوير مواقع الإنترنت، بالإضافة إلى إمكانية العمل في شركات برمجيات محلية أو عالمية.

قطاع الاتصالات

العمل في شركات الاتصالات وتطوير الأنظمة الرقمية وخدمات العملاء الإلكترونية.

الوزارات والهيئات الحكومية

شغل وظائف حكومية بمجالات التحول الرقمي، إدارة قواعد البيانات، وصيانة وتطوير الأنظمة المعلوماتية.

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سياسة الخصوصية

روابط مهمة

الموقع الرسمي لجامعة أسيوط
الموقع الرسمي لكلية الحاسبات والمعلومات

للتواصل

الهاتف: ٨٨٢٤٣٩٦
البريد الإلكتروني: pg_fci@aun.edu.eg

Calculating the download time for the webpage

1-Calculate the size of the HTML file and all its associated images in kiloByte.

2-Determine the speed of your network connection. Some of the speeds are

28.8 Kbps modem speed

33.6 Kbps modem speed

56 Kbps modem speed

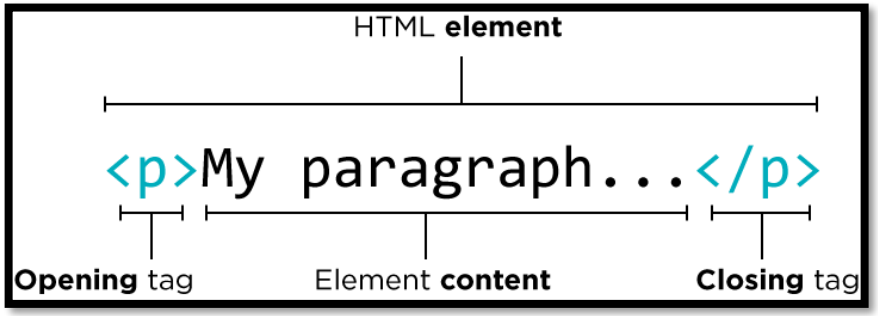
128 , 256, 512 Kbps, 1, 2 Mbps DSL or ADSL line

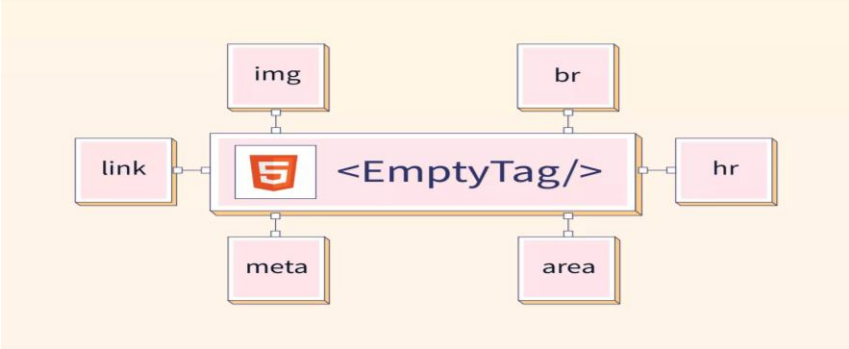
Divide the total size by the speed to get the download time.

Creating HTML Page

Tags and Elements

- Containers Tags
- Empty Tags
  - <img>: Represents an image.
  -
: Inserts a line break.
  - <hr>: Inserts a horizontal rule.
  - <input>: Creates an input field for various types of data.
  - <meta>: Provides metadata about the HTML document.
  - <link>: Links to external resources, such as stylesheets or scripts.





The Structure of a Web Page

```html
<!DOCTYPE html>
<html lang="en">
<head>
    <meta charset="UTF-8">
    <meta name="viewport" content="width=device-width, initial-scale=1.0">
    <title>Web Page Title</title>
    <!-- CSS and Scripts -->
    <link rel="stylesheet" href="styles.css">
    <script src="script.js"></script>
</head>
<body>
    <!-- Web Page Content -->
</body>
</html>
```

Common Body Sections and Elements

- Header (<header>):

```
<header>
    <h1>Website Name</h1>
    <nav>
        <ul>
            <li><a href="#home">Home</a></li>
            <li><a href="#about">About</a></li>
            <li><a href="#contact">Contact</a></li>
        </ul>
    </nav>
</header>
```

Common Body Sections and Elements

- Navigation (<nav>): Holds links for navigating through the website. Typically included in the header or in a sidebar.
- Main Content (<main>):Contains the primary content of the page. This section should have a clear purpose and structure, usually divided into subsections, articles, and sections.
- Aside (<aside>):Contains additional information, such as sidebars, quotes, or advertisements.
- Footer (<footer>): Contains footer content like copyright information, social media links, or contact details.

Common Body Sections and Elements (the page)

Welcome to My Website

A place to showcase common web page sections and elements

Home About Services Contact

Home

Welcome to the homepage! This is where the main content of the web page goes.

Lorem ipsum dolor sit amet, consectetur adipiscing elit. Nullam ut magna et nulla facilisis pretium. Praesent malesuada libero sit amet risus aliquam, sed tincidunt eros consequat.

About

This section contains information about the website or the individual it represents.

Lorem ipsum dolor sit amet, consectetur adipiscing elit. Nullam sit amet pulvinar lorem. Nam in risus non sapien dignissim tincidunt. Cras nec magna ut arcu bibendum posuere.

Sidebar

This sidebar can contain additional information, links, advertisements, or other content not central to the main area.

- Related Link 1
- Related Link 2
- Related Link 3

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Components of an HTML Tag

- A Tag Consists of Three Items:
  - An Element: Represents the HTML structure. Example: <TITLE>, <BODY>, <DIV>, etc.
  - An Attribute: Specifies a certain aspect of the element. Example:<DIV> has attributes like align, class, id, etc.
  - A Value: Provides a value for the attribute. Example:<DIV align="center">
- Note
  - Some elements have attributes, while others do not.

Document Structure Tags

At a minimum , an HTML 4.01 document must contain the following tags in the sequence shown

```
<!DOCTYPE HTML PUBLIC “-//W3C//DTD HTML 4.01 Transitional//EN”
    “http://www.w3.org//TR//html14/loose.dtd”>

<HTML>
    <HEAD>
        <TITLE>Descriptive page</TITLE>
    </HEAD>
    <BODY>
        Page text and media
    </BODY>
</HTML>
```

The Document type declaration (DTD) begins the HTML code by specifying the HTML version and type

HTML Attribute Groups

Global Attributes

- Attributes that can be applied to any HTML element.
- Examples:
  - class: Specifies one or more class names for an element.
  - id: Specifies a unique identifier for an element.
  - style: Applies inline CSS styles to an element.

Form Attributes

- Attributes specific to form elements.
- Examples:
  - action: Specifies the URL to send form data.
  - method: Defines the HTTP method to use (GET or POST).
  - placeholder: Provides a hint to the user of what can be entered in the field.

Event Attributes

- Attributes that trigger JavaScript events.
- Examples:
  - onclick: Executes a script when the element is clicked.
  - onmouseover: Executes a script when the mouse hovers over the element.

Media Attributes

- Attributes used with media elements (like <audio>, <video>).
- Examples:
  - controls: Displays playback controls.
  - autoplay: Automatically plays the media when loaded.
  - loop: Plays the media in a loop.

The id Attribute id="*string*"

- The id attribute can be used to uniquely identify any element within a page or style sheet. There are two primary reasons that you might want to use an id attribute on an element.
  - If an element carries an id attribute as a unique identifier it is possible to identify just that element and its content (perhaps you want to link to a specific point in a document, to select one specific element's content, to associate a CSS style
  - If you have two elements of the same name within a Web page or style sheet, you can use the id attribute to distinguish between elements that have the same name

The title Attribute title="*string*“

- it is often displayed as a **tooltip** or while the element is loading.

The class Attribute class="className“ The class attribute is used to associate an element with a style sheet .

- ***Internationalization attributes***

  There are three internationalization attributes, dir lang xml:lang

The dir Attribute

- **Value Meaning**

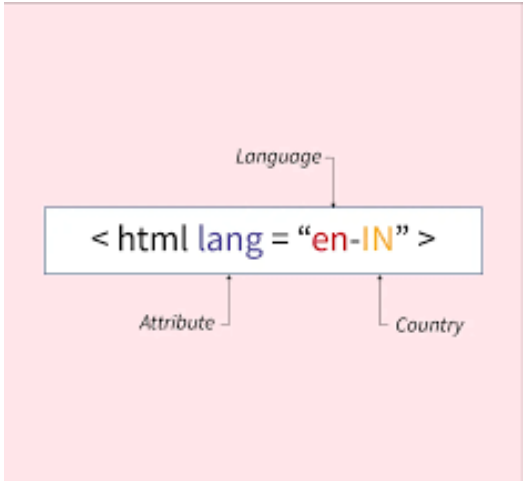
ltr Left to right (the default value)

rtl Right to left (for languages such as Arabic that are read right to left).

The lang Attribute

- **Value** **Meaning**

ar	Arabic
en	English
en-us	U. S. English
zh	Chinese



HTML
dir
Attribute

The xml:lang attribute is the XHTML replacement for the lang.

Extra UI Events Attributes

- onclick, ondoubleclick, onmousedown, onmouseup, onmouseover, onmousemove, onmouseout, onkeypress, onkeydown, onkeyup

Basic Text Formatting in HTML

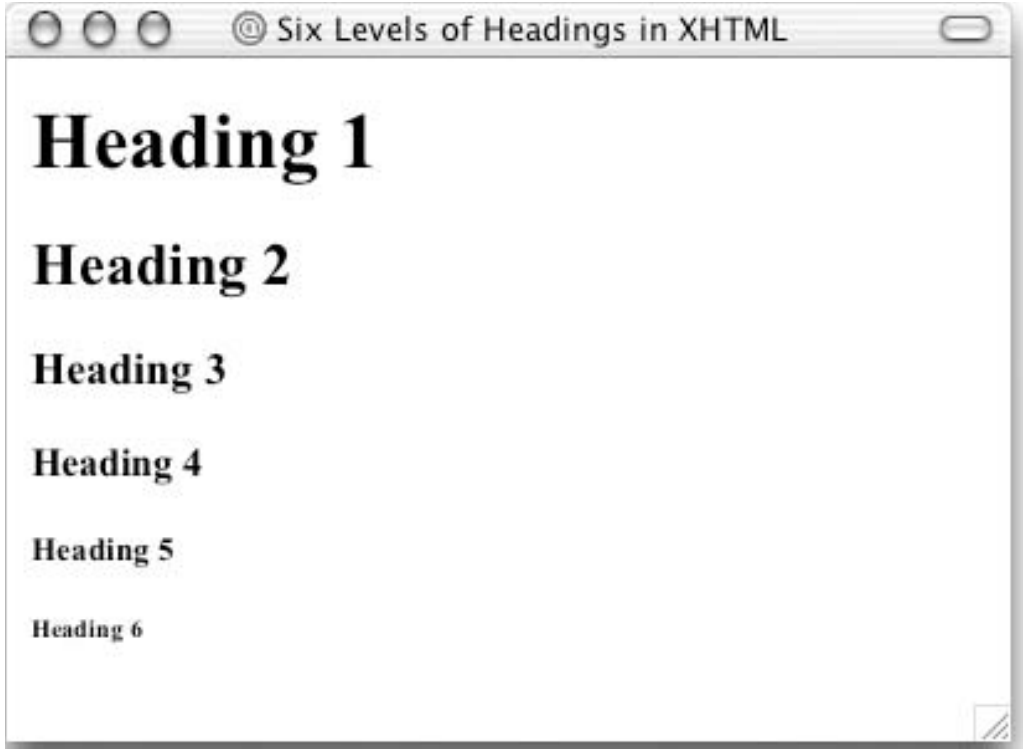
Headings
•Define headings of different levels.
•Tags: <h1> to <h6>

Paragraphs
•Defines a block of text.
•Tag: <p>

Text Styles
•**Bold Text**: or <b>
    •Example: Bold Text
•**Italic Text**: or <i>
    •Example: Italic Text
•**Underlined Text**: <u>
    •Example: <u>Underlined Text</u>

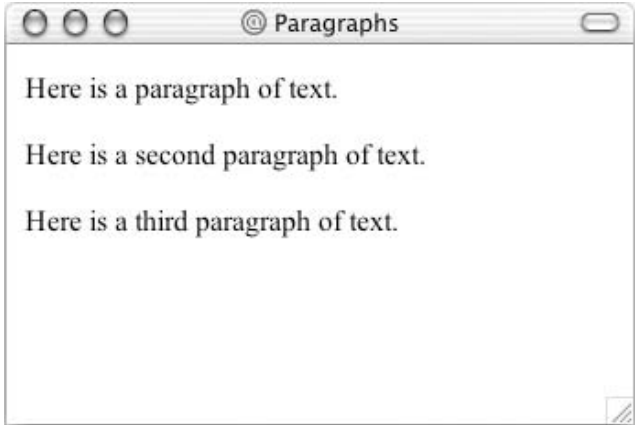
Lists
•**Unordered List**: <ul> with <li>

Creating Headings Using hn Elements



Creating Paragraphs Using the Element

- Here is a paragraph of text.
- Here is a second paragraph of text.
- Here is a third paragraph of text.



***Creating Line Breaks Using the
 Element***

Paragraph one

Paragraph two

It is used to push text down the page

- ***Creating Preformatted Text Using the <pre> Element.***
- Any text between the opening <pre> tag and the closing </pre> tag will preserve the formatting of the source document.

Presentational Elements

- *The <b> Element*

Anything that appears in a <b> element is displayed in **bold**

- *The <i> Element*

The content of an <i> element is displayed in *italicized* text

- *The <u> Element (deprecated)*

The content of a <u> element is *underlined*

- ***The <tt> Element***

  The content of a <tt> element is written in *monospaced* font.

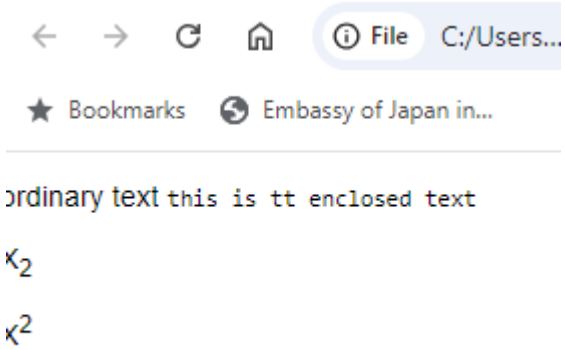
- ***The <sup> Element***

  The content of a <sup> element is written in *superscript.*

  Written on the 31st February.

- ***The <sub> Element***

  The content of a <sub> element is written in subscript

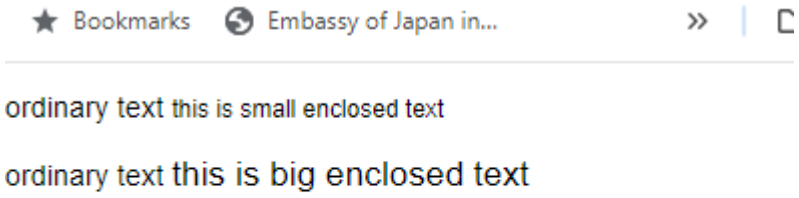


- ***The <big> Element***

The content of the <big> element is displayed one font size larger than the rest of the text surrounding it.

- ***The <small> Element***

The content of the <small> element is displayed one font size smaller than the rest of the text.



Lists

There are many reasons why you might want to add a list to your page.

You can create three types of lists in XHTML:

- **Unordered lists**, which are like lists of bullet points
- **Ordered lists**, which use a sequence of numbers or letters instead of bullet points
- **Definition lists**, which allow you to specify a term and its definition

- ***Using the <ul> Element to Create Unordered Lists***
- Each bullet point or line you want to write should then be contained between opening <li> tags and closing </li> tags.

```
<ul>
<li>Bullet point number one</li>
<li>Bullet point number two</li>
<li>Bullet point number three</li>
</ul>
```

- ***Ordered Lists***
- Sometimes, you want your lists to be ordered. In an ordered list, rather than prefixing each point with a bullet point, you can use either numbers (1, 2, 3), letters (A, B, C) or Roman numerals (i, ii, iii) to prefix the list item.
- **Value for type Attribute Description Examples**

1 Arabic numerals (the default) 1, 2, 3, 4, 5

A Capital letters A, B, C, D, E

a Small letters a, b, c, d, e

I Large Roman numerals I, II, III, IV, V

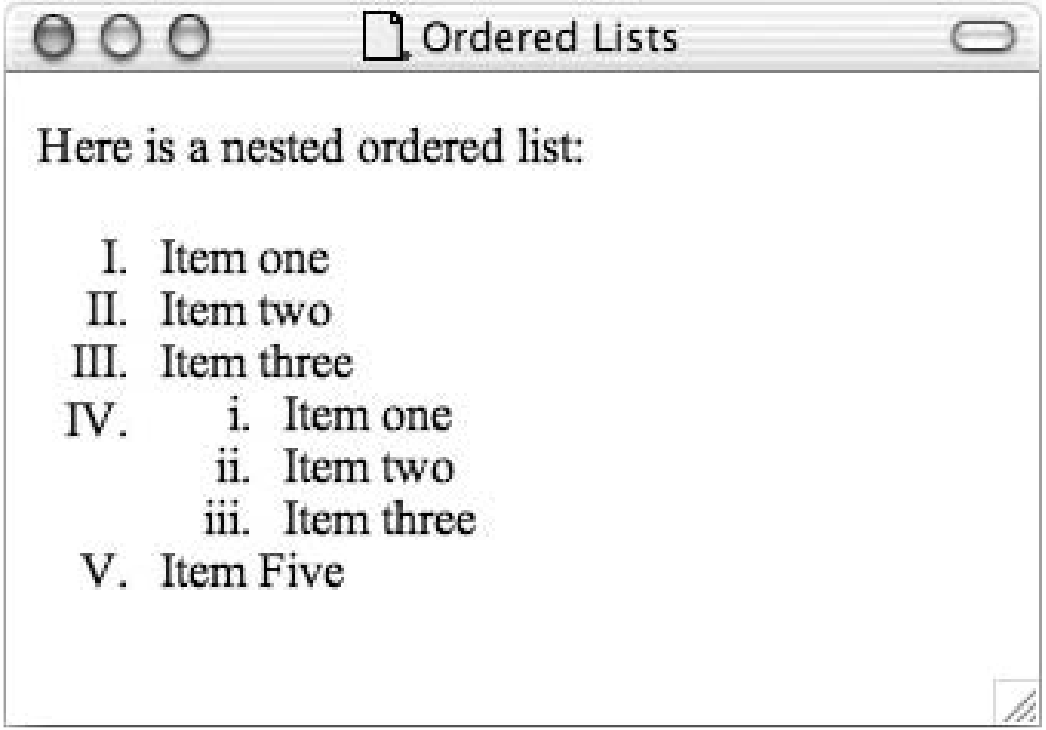
i Small Roman numerals i, ii, iii, iv, v

- ***Definition Lists***

```
<dl>
<dt>Unordered List</dt>
<dd>A list of bullet points.</dd>
<dt>Ordered List</dt>
<dd>An ordered list of points, such as a numbered set of
    steps.</dd>
<dt>Definition List</dt>
<dd>A list of terms and definitions.</dd>
</dl>
```

- ***Nesting Lists***

```
<ol type="I">
<li>Item one</li>
<li>Item two</li>
<li>Item three</li>
<li>
<ol type="i">
<li>Item one</li>
<li>Item two</li>
<li>Item three</li>
</ol>
</li>
<li>Item Five</li>
</ol>
```



- Comments

You can put comments between any tags in your XHTML documents.

Comments use the following

syntax:

<!-- comment goes here -->

Anything after <!-- until the closing --> will not be displayed.

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Comment

Safari File Edit View History Bookmarks Develop Window Help

w3schools.com

Run ›

Result Size: 561 x 476

Get your own website

```
<!DOCTYPE html>
<html>

    <head>
        <title>
            Hello World Wide Web!
        </title>
    </head>

    <body>
        <h1>
            Display this as a heading h1.
        </h1>

        <h3>
            The content of the body element is displayed in
your browser.
        </h3>
        <p>
            This is a paragraph
        </p>
        <! This is a comment! >
    </body>

</html>
```

Display this as a heading h1.

The content of the body element is displayed in your browser.

This is a paragraph

Links and Navigation

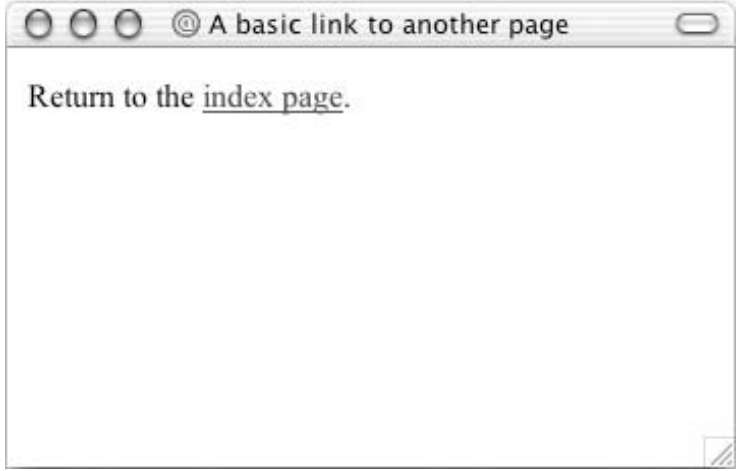
- How to structure the folders on your Web site.
- How to link between pages of your site.
- How to link to specific parts of a page in your site.
- How to link to other sites.

Basic Links

- A link is specified using the <a> element. Anything between the opening <a> tag and the closing </a> tag becomes part of the link a user can click in a browser.

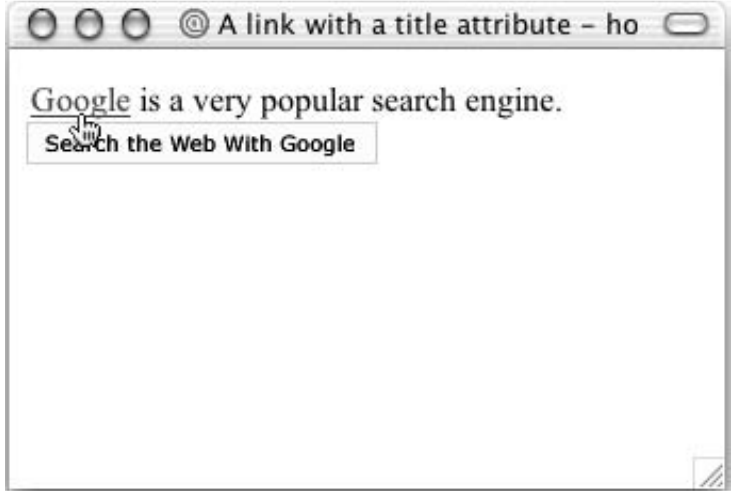
Linking to Other Documents

```
<body>
Return to the <a href="index.html">index
    page</a>.
</body>
```



A link could be to a site

```
<a href="http://www.Google.com/" title="Search the
   Web with Google">Google</a>
is a very popular search engine.
```



Linking to E-mail Addresses

```
<a href="mailto:name@example.com">E-mail
    us</a>.
```

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Links & White Spaces

```
<A href="https://www.w3schools.com/">Go to W3Schools</A>
<A href="https://www.stmartin.edu/">Go to St. Martin University</A>
```

```
<A href="firstwebpage.html">Go to the first page</A>
```

Display this as another heading

Go to W3Schools Go to St. Martin University Go to the first page

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Links & White Spaces

```
<A href="https://www.w3schools.com/">Go to W3Schools</A>
<A href="https://www.stmartin.edu/">Go to St. Martin University</A>
```

link

text

```
<A href="firstwebpage.html">Go to the first page</A>
```

Display this as another heading

Go to W3Schools Go to St. Martin University Go to the first page

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Links & White Spaces (trim and Subress)

```
<A href="https://www.w3schools.com/">Go to W3Schools</A>
<A href="https://www.stmartin.edu/">Go to St. Martin University</A>

<A href="firstwebpage.html">Go to the first page</A>
```

Space???

Display this as another heading

Go to W3Schools Go to St. Martin University Go to the first page

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White Spaces

Safari File Edit View History Bookmarks Develop Window Help 21% Sat 16 Sep 4:38 PM

w3schools.com

Run ›

Result Size: 561 x 471

Get your own website

```
<!DOCTYPE html>
<html>

    <head>
        <title>
            Hello World Wide Web!
        </title>
    </head>

    <body>

         <A href="www.w3schools.com"> Go to W3 School </A>


         <A href="www.google.com"> Go to Google </A>

    </body>

</html>
```

Go to W3 School Go to Google

© BIS401

White Spaces

Safari File Edit View History Bookmarks Develop Window Help 18% Sat 16 Sep 4:35 PM

w3schools.com

Run ›

Result Size: 561 x 471

Get your own website

```
<!DOCTYPE html>
<html>

    <head>
        <title>
            Hello World Wide Web!
        </title>
    </head>

    <body>

         <A href="www.w3schools.com"> Go to W3 School </A>

         <BR>


         <A href="www.google.com"> Go to Google </A>

    </body>

</html>
```

Go to W3 School
Go to Google

© BIS401

White Spaces

Safari File Edit View History Bookmarks Develop Window Help 18% Sat 16 Sep 4:35 PM

w3schools.com

Run ›

Result Size: 561 x 471

Get your own website

```
<!DOCTYPE html>
<html>

    <head>
        <title>
            Hello World Wide Web!
        </title>
    </head>

    <body>

         <A href="www.w3schools.com"> Go to W3 School </A>

         <BR>
         <BR>

         <A href="www.google.com"> Go to Google </A>

    </body>

</html>
```

Go to W3 School

Go to Google

© BIS401

White Spaces

Safari File Edit View History Bookmarks Develop Window Help 18% Sat 16 Sep 4:36 PM

w3schools.com

Run ›

Result Size: 561 x 471

Get your own website

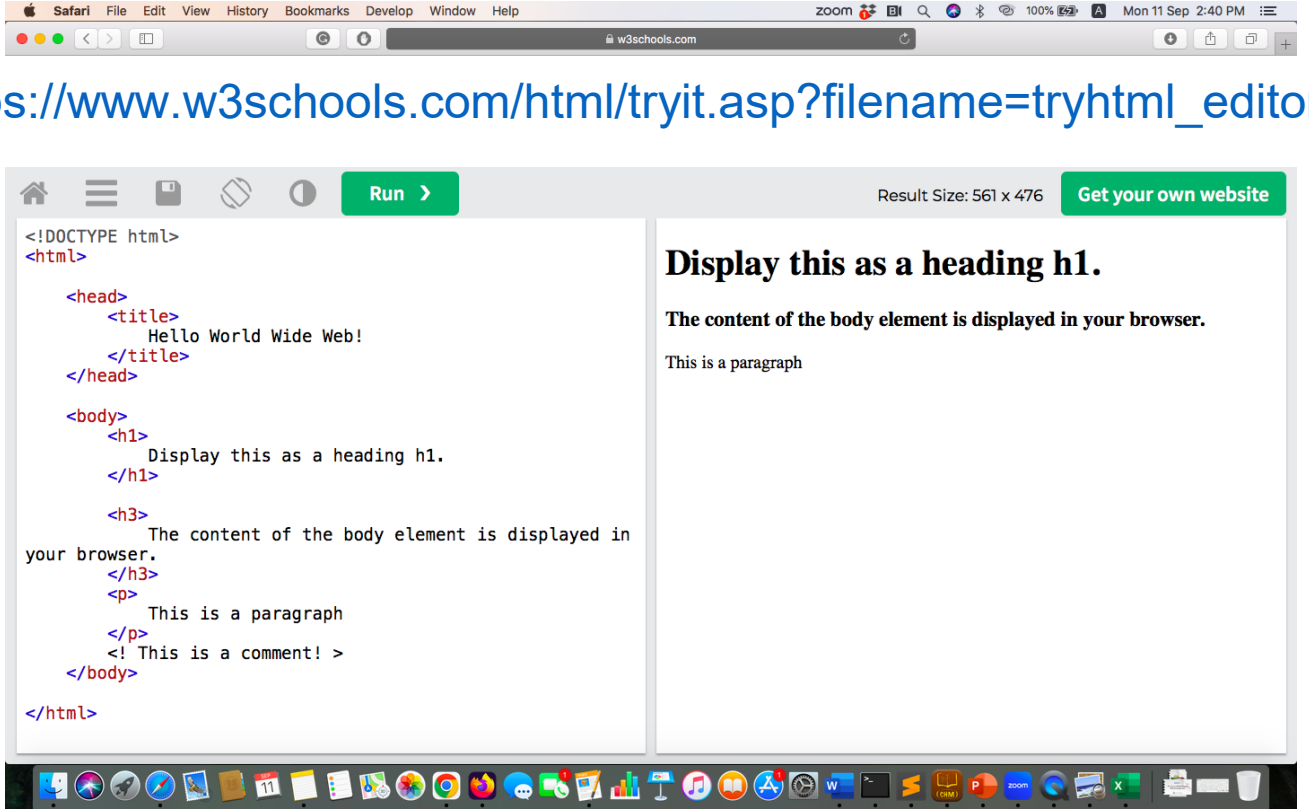
```
<!DOCTYPE html>
<html>

    <head>
        <title>
            Hello World Wide Web!
        </title>
    </head>

    <body>

         <A href="www.w3schools.com"> Go to W3 School </A>

         <BR>
         <BR>
         <BR>

         <A href="www.google.com"> Go to Google </A>

    </body>

</html>
```

Go to W3 School

Go to Google

© BIS401

Tools: A First HTML Document



https://www.w3schools.com/html/tryit.asp?filename=tryhtml_editor

Development Environment

A Web Browser:

Google Chrome browser ⇒ https://www.google.com/ chrome/

Others: Mozilla Firefox, Safari, …

A Text Editor:

Visual Studio Code (VS Code) ⇒ https://visualstudio.microsoft.com/

Others: Vim, Notepad, Sublime, …

Development Environment (Back End)

C# Compiler:

Visual Studio Community 2022 ⇒ https://visualstudio.microsoft.com/

Database Framework:

DB Browser for SQLite ⇒ https://sqlitebrowser.org/dl/

Development Environment: Optional

Show File Extension:

.html, .js, …

Database Framework:

Microsoft SQL Server ⇒ https://aka.ms/ssmsfullsetup

Sample Data Generator:

Fake Name Generator ⇒ https://commentpicker.com/fake-name-generator.php

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Colors, Images, and Objects

Adding Color to your site

In XHTML there are two key ways of specifying a color:

1- **Hex codes:** A six-digit code representing the amount of red, green, and blue that make up the
color, preceded by a pound or hash sign # (for example, #333333).

2- **Color names:** A set of names that represent over 200 colors, such as red, cyan, purple,…

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Using Hex Codes to Specify Colors

Color	Hexadecimal Code
Black	#000000
White	#FFFFFF
Red	#FF0000
Green	#008000
Blue	#0000FF
Purple	#800080

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Using Color Names to Specify Colors

There are over 200 different color names supported by Netscape and IE, names might sound a lot easier to understand than hex codes,

aqua, beige, coral, darkcyan, firebrick, green, honeydew, indianred, lavenderblush, maroon, navy, oldlace, palegreen, red, saddlebrown, tan, white, yellow

Color has a very strong influence on people and can affect how people view a company. So, when designing any Web site, color should be an important factor. A well-chosen color scheme is key to making any site attractive.

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The Basics of Color

There are three *primary colors*: red, yellow, and blue.

Combining primary colors gives you *secondary colors* of green, orange, and purple.

Tertiary colors are a combination of primary and secondary colors: red-orange, red-purple, blue-purple, blue-green, yellow-orange, and yellow green.

These are 12 colors

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Designer use other terminology such as *hue, shade, tint, and intensity*

A *hue* is a pure color; it contains no black or white. It is the key part of a color that allows it to be identified as red, green, or blue.

A *shade* is a hue with black added.

A *tint* is a hue with white added.

A *tone* is a hue with gray added.

Intensity describes the intensity of a color, sometimes the terms *saturation* or *chromaticity* are used

The Web Safe Color Palette

- Web safe colors are a set of 216 colors that were widely used in the early days of web design to ensure consistent color display across different computer monitors, particularly when many displays were limited to 256 colors. These colors were created using a combination of six levels for each of the RGB (Red, Green, Blue) color values: 00, 33, 66, 99, CC, and FF.

- This palette was designed so that colors would display consistently regardless of the browser or device, but modern displays and browsers can show millions of colors, so web safe colors are less critical today.

Some WSCP

Adding Images to Your Site

- Images and graphics can really bring your site to life.
- You should be very careful when using images in your site as they slow down the speed of site downloading. You should choose image format and sizes very carefully to avoid slow down problems.

Types of Image Formats

- *Bitmapped graphics* divide a picture into a grid of pixels and specify the color of each pixel, There are several different formats of bitmap; common ones include (the rather confusingly named) bitmap or BMP, JPEG, GIF, TIFF, and PNG. You will be learning more about JPEGs, GIFs, and PNGs

- *Vector graphics* break the image into lines and shapes (like a wireframe drawing), and store the lines as coordinates. They then fill the spaces between the lines with color. Vector graphics are commonly used for line art, illustration, and animation.

Bitmap Images

- Most static images on the Web will be bitmapped images. Browsers tend to support three common bitmap graphics formats, and most graphics programs will save images in these formats:
- **GIF:** Graphics Interchange format (pronounced either “gif” or “jif”)
- **JPEG:** Joint Photographic Experts Group format (pronounced “jay peg”)
- **PNG:** Portable Network Graphics (pronounced “ping” or “pee en gee”)

GIF Images

- The GIF (or Graphics Interchange Format) used to be the standard for all Web graphics. GIF is an *indexed color format* that allows an image to select up to 256 colors from a range of over 16 million colors to recreate the picture; each pixel is one of these 256 colors

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JPEG Images

- The JPEG image format was developed as a standard for storing and compressing images such as

photographs with wide ranges of colors.

- PNG Images

The Portable Network Graphics format is the most recent format on the block. It was developed in the

- late 1990s because the company that owns the patent for GIFs (Unisys) decided to charge companies that developed software for creating and viewing GIFs a license fee to use the technolog

Keeping File Sizes Small

- Try to use the smaller size images in order to download fast.

- Use JPEGs for photo-realistic pictures with a lot of detail, or subtle shade differences you want to preserve.
- Use GIFs for images with flat color (rather than textured colors), and hard edges, such as diagrams, text, or logos.

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Adding Images Using the <img> Element.

<img src="logo.gif" alt="Wrox logo"/>

Attributes

Src, alt, align, border, height, width, hspace, vspace, ismap, usemap longdesc, name

Source Attribute

src="url“

The alt Attribute

- is required to specify a text alternative for the image in case the browser cannot display the image.

alt=“FCI logo“

The align Attribute (deprecated)

align="right“

top :The top of the image is aligned with top of the current line of text.

middle, bottom, left, right

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- The border Attribute (deprecated)

The border attribute specifies the width of the border around the image in pixels:

border="2“

- The height and width Attributes

The height and width attributes specify the height and width of the image:

height="120" width="180“

- The hspace and vspace Attributes (deprecated)
- The hspace and vspace attributes are used to control the amount of whitespace around an image.
- hspace="10"
- vspace="14“

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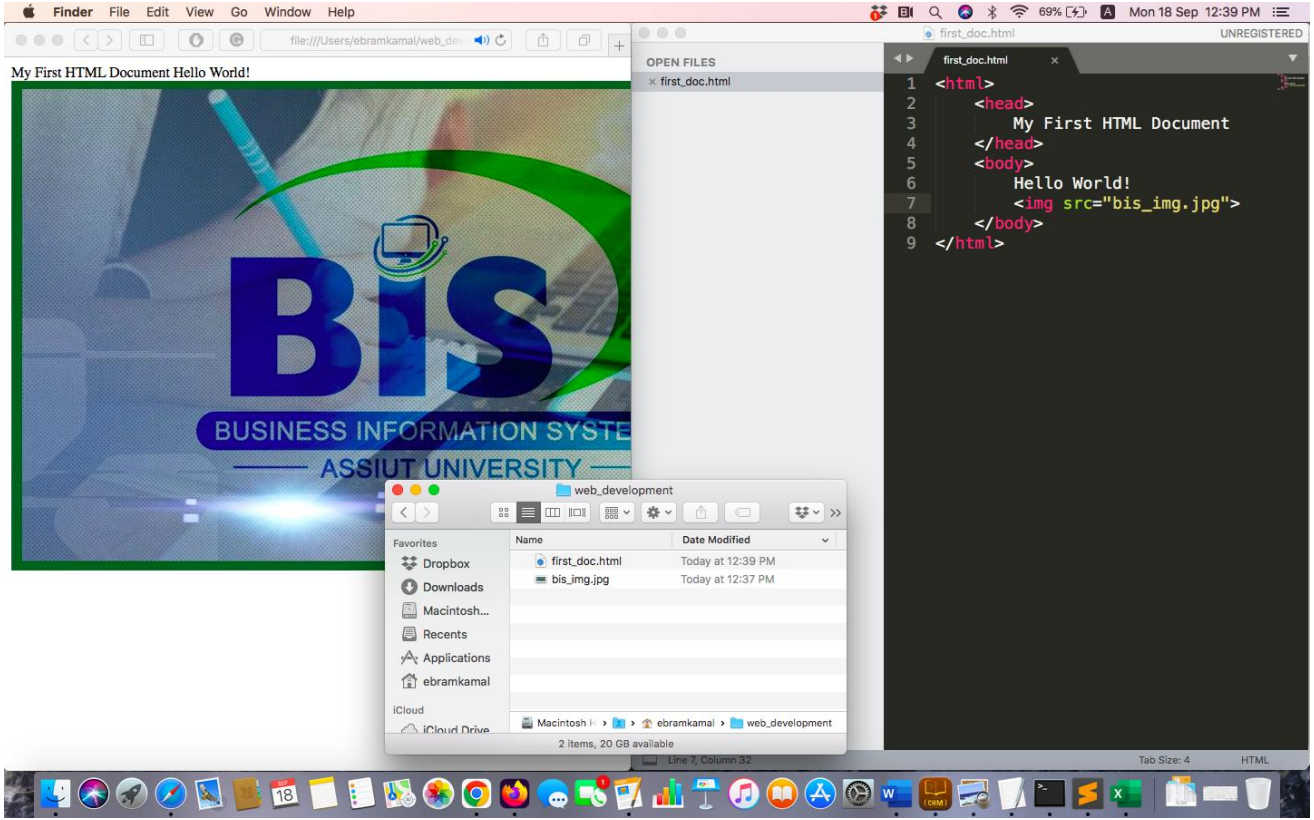
Images



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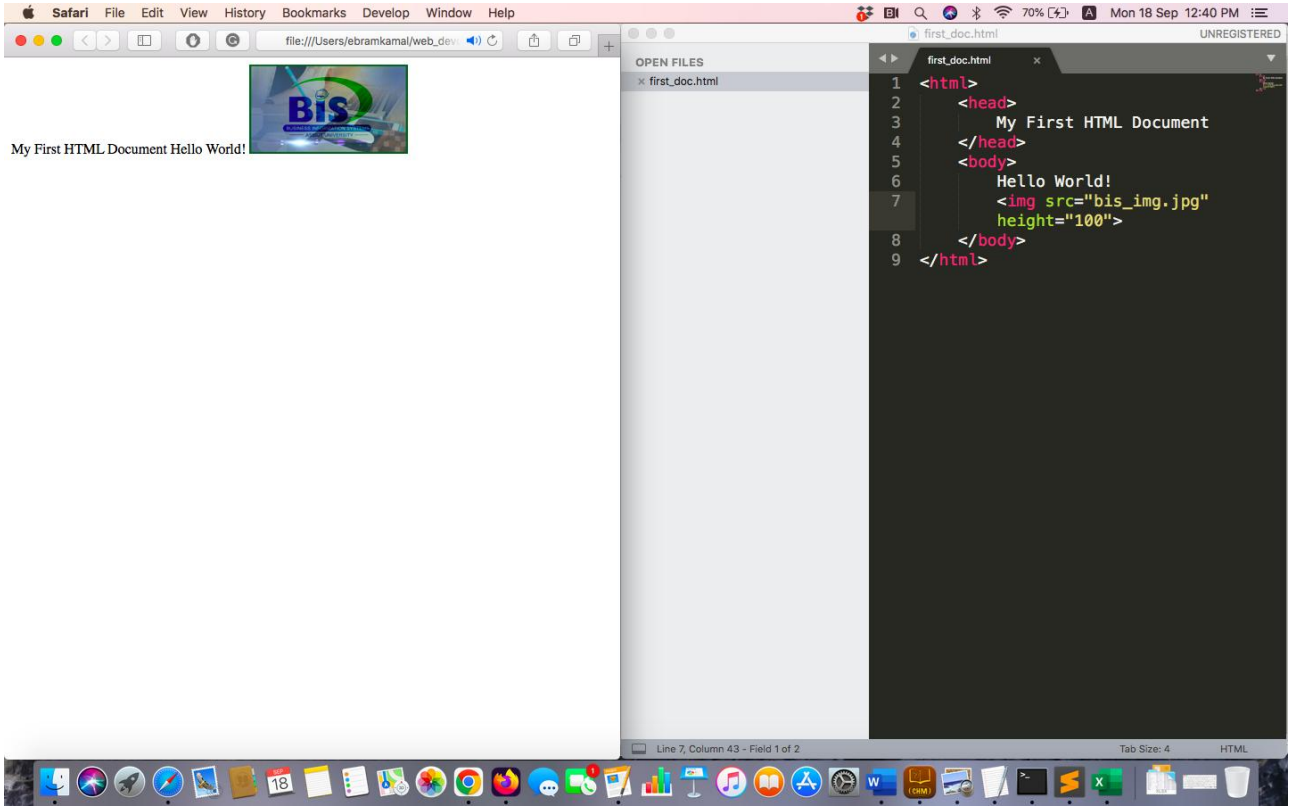
Images



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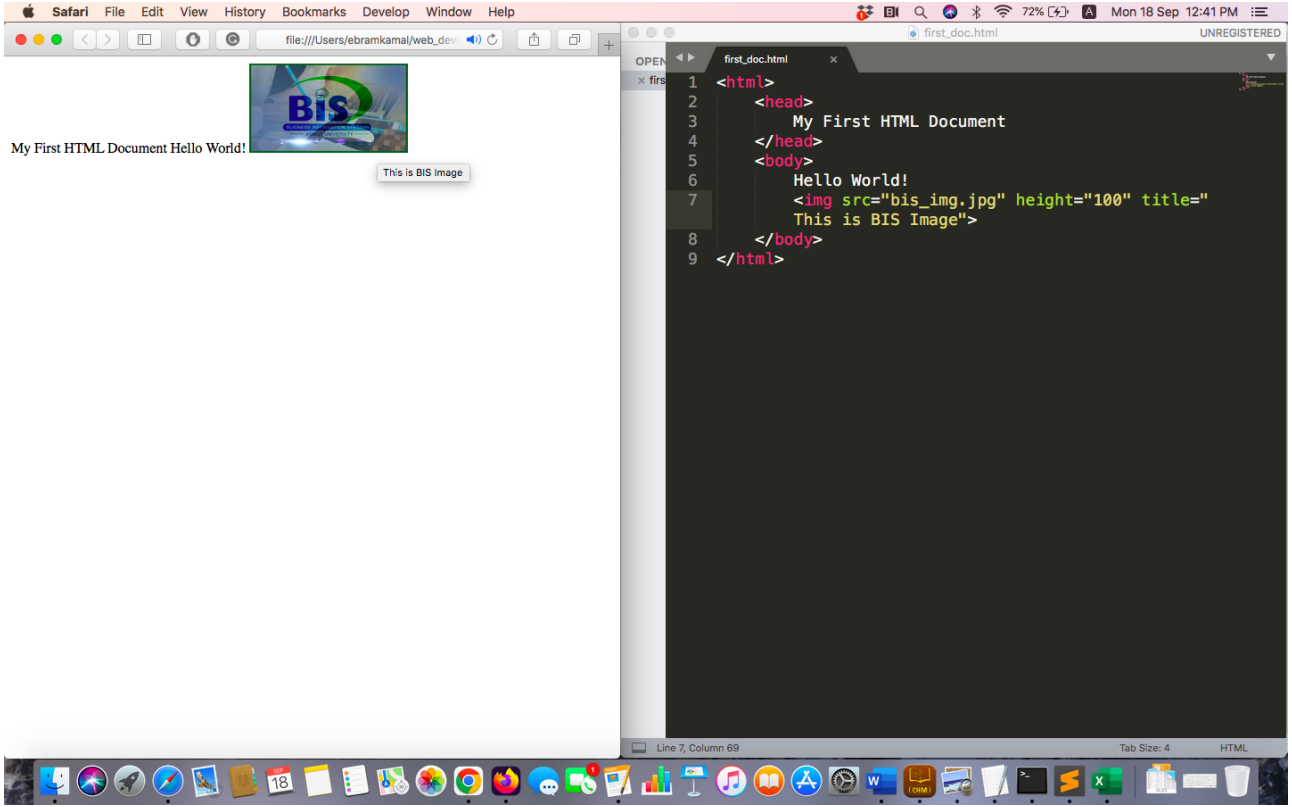
Images



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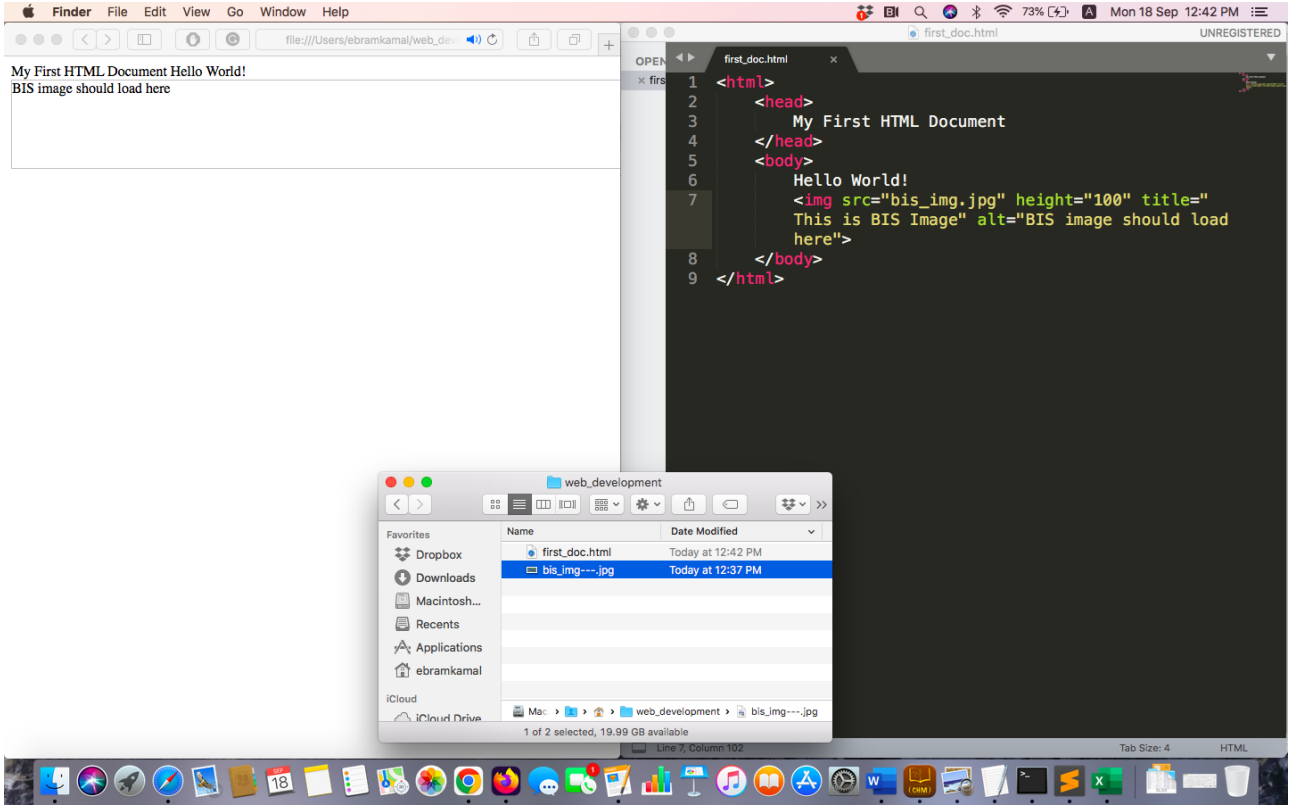
Images



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Tables

Current Enrollment for Razvan Mezei

Course ID	Course name	Course link
CSC200	Object Oriented Programming	Course link
CSC340	Data Structures and Algorithms	Course link
CSC495	ST ASP .Net Core MVC	Course link

Go to W3Schools Go to St. Martin University Go to the first page

```
<TABLE>
    <THEAD>
        <TR>
            <TH>Course ID</TH>
            <TH>Course name</TH>
            <TH>Course link</TH>
        </TR>
    </THEAD>
    <TBODY>
        <TR>
            <TD>CSC200</TD>
            <TD>Object Oriented Programming</TD>
            <TD><a href="https://www.stmartin.edu/">Course link</a></TD>
        </TR>
        <TR>
            <TD>CSC340</TD>
            <TD>Data Structures and Algorithms</TD>
            <TD><a href="https://www.stmartin.edu/">Course link</a></TD>
        </TR>
        <TR>
            <TD>CSC495</TD>
            <TD>ST ASP .Net Core MVC</TD>
            <TD><a href="https://www.stmartin.edu/">Course link</a></TD>
        </TR>
    </TBODY>
</TABLE>
<BR>
<BR>
```

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Tables

- In order to work with tables, you need to start thinking in *grids*. Tables, just like spreadsheets, are made up of rows and columns, as shown in Figure

```
<table border="1">
<tr>
<td>Row 1, Column 1</td>
<td>Row 1, Column 2</td>
</tr>
<tr>
<td>Row 2, Column 1</td>
<td>Row 2, Column 2</td>
</tr>
</table>
```

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Rows

Columns

Column 1 Row 1	Column 2 Row 1	Column 3 Row 1	Column 4 Row 1	Column 5 Row 1
Column 1 Row 2	Column 2 Row 2	Column 3 Row 2	Column 4 Row 2	Column 5 Row 2
Column 1 Row 3	Column 2 Row 3	Column 3 Row 3	Column 4 Row 3	Column 5 Row 3
Column 1 Row 4	Column 2 Row 4	Column 3 Row 4	Column 4 Row 4	Column 5 Row 4
Column 1 Row 5	Column 2 Row 5	Column 3 Row 5	Column 4 Row 5	Column 5 Row 5

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Table headings have their own element, <th>, which is used in place of the table data or <td> element.
The <th> element can be used for either column or row headers.

```
<table border="1">
<tr>
<th></th>
<th>Outgoings ($)</th>
<th>Receipts ($)</th>
<th>Profit ($)</th>
</tr>
<tr>
```

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```
<th>Quarter 1 (Jan-Mar)</th>
<td>11200.00</td>
<td>21800.00</td>
<td><b>10600.00</b></td>
</tr>
<tr>
<th>Quarter 2 (Apr-Jun)</th>
<td>11700.00</td>
<td>22500.00</td>
<td><b>10800.00</b></td>
</tr>
<tr>
<th>Quarter 3 (Jul - Sep)</th>
<td>11650.00</td>
<td>22100.00</td>
<td><b>10450.00</b></td>
</tr>
<tr>
<th>Quarter 4 (Oct - Dec)</th>
<td>11850.00</td>
<td>22900.00</td>
<td><b>11050.00</b></td>
</tr>
</table>
```

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- The <table> element can carry the following deprecated attributes.
- Even though they are deprecated, you will still see many of them in use today:

align bgcolor border cellpadding cellspacing dir frame rules summary width

The <td> and <th> Elements Represent Table Cells

```
<table border="1">
  <tr>
    <th>First Name</th>
    <th>Last Name</th>
    <th>Age</th>
  </tr>
  <tr>
    <td>John</td>
    <td>Doe</td>
    <td>30</td>
  </tr>
  <tr>
    <td>Jane</td>
    <td>Smith</td>
    <td>25</td>
  </tr>
</table>
```

First Name	Last Name	Age
John	Doe	30
Jane	Smith	25

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- *Attributes of the <tr> and <td>*
- *abbr align axis bgcolor char charoff colspan headers height nowrap rowspan scope valign width abbr Attribute*

Advanced Tables

Let's explore these more advanced techniques for working with tables in HTML:

1. Splitting a Table into Three Sections: <thead>, <tbody>, and <tfoot>

- You can organize a table into three distinct parts:
- <thead>: Contains the table header.
- <tbody>: Contains the main content of the table.
- <tfoot>: Contains the footer, often used for summaries or totals.

```
<table border="1">
  <thead>
    <tr>
      <th>Item</th>
      <th>Price</th>
    </tr>
  </thead>
  <tbody>
    <tr>
      <td>Apple</td>
      <td>$1.00</td>
    </tr>
    <tr>
      <td>Banana</td>
      <td>$0.50</td>
    </tr>
  </tbody>
  <tfoot>
    <tr>
      <td>Total</td>
      <td>$1.50</td>
    </tr>
  </tfoot>
</table>
```

2. Captioning Tables Using the <caption> Element

- The <caption> element adds a descriptive title to the table, which is typically displayed above the table.

```
<table border="1">
  <caption>Fruit Prices</caption>
  <thead>
    <tr>
      <th>Item</th>
      <th>Price</th>
    </tr>
  </thead>
  <tbody>
    <tr>
      <td>Apple</td>
      <td>$1.00</td>
    </tr>
    <tr>
      <td>Banana</td>
      <td>$0.50</td>
    </tr>
  </tbody>
</table>
```

Fruit Prices

Item	Price
Apple	$1.00
Banana	$0.50

Using <tfoot>

3. Using rowspan and colspan Attributes

- rowspan allows a cell to span multiple rows.
- colspan allows a cell to span multiple columns.

Item	Price Info	
Apple	Retail	$1.00
	Wholesale	$0.80

```
<table border="1">
  <thead>
    <tr>
      <th>Item</th>
      <th colspan="2">Price Info</th>
    </tr>
  </thead>
  <tbody>
    <tr>
      <td rowspan="2">Apple</td>
      <td>Retail</td>
      <td>$1.00</td>
    </tr>
    <tr>
      <td>Wholesale</td>
      <td>$0.80</td>
    </tr>
  </tbody>
</table>
```

4. Grouping Columns Using the <colgroup> Element

- The <colgroup> element allows you to group columns, often for styling or other purposes like setting widths.

Item	Price
Apple	$1.00
Banana	$0.50

```
<table border="1">
  <colgroup>
    <col style="background-color:lightgray">
    <col style="background-color:lightblue">
  </colgroup>
  <thead>
    <tr>
      <th>Item</th>
      <th>Price</th>
    </tr>
  </thead>
  <tbody>
    <tr>
      <td>Apple</td>
      <td>$1.00</td>
    </tr>
    <tr>
      <td>Banana</td>
      <td>$0.50</td>
    </tr>
  </tbody>
</table>
```

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Select

```
<SELECT id="majors">
    <OPTION value="Art">Art</OPTION>
    <OPTION value="Math">Mathematics</OPTION>
    <OPTION value="CS">Computer Science</OPTION>
    <OPTION value="Undecided">Undecided</OPTION>
</SELECT>
```

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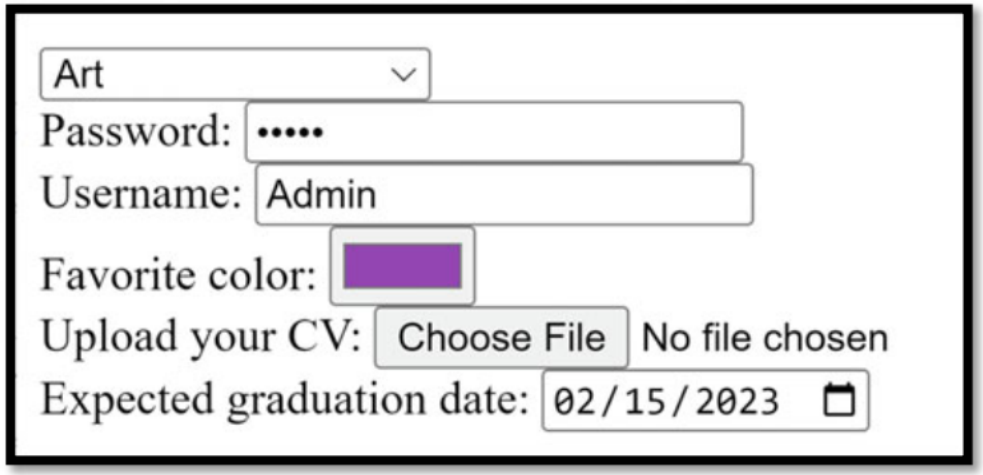
Label & Input

```
<LABEL for="passw">Password: </LABEL> <INPUT type="password" id="passw"> <BR>
<LABEL for="user">Username: </LABEL> <INPUT type="text" id="user"> <BR>
<LABEL for="color">Favorite color: </LABEL> <INPUT type="color" id="color"> <BR>
<LABEL for="cv">Upload your CV: </LABEL> <INPUT type="file" id="cv"> <BR>
<LABEL for="gd">Expected graduation date: </LABEL> <INPUT type="date" id="gd"> <BR>
```



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Form

```
<FORM>
    <LABEL for="passw">Password: </LABEL>
    <INPUT type="password" id="passw" name="password" required>
    <BR>
    <LABEL for="user">Username: </LABEL>
    <INPUT type="text" id="user"  name="username" placeholder="choose a username">
    <BR>
    <LABEL for="email">Email address: </LABEL>
    <INPUT type="email" id="email"  name="email">
    <BR>
    <LABEL for="phone">Phone number: </LABEL>
    <INPUT type="tel" id="phone"  name="phone" value="000-000-0000">
    <BR>
    <LABEL for="veteran">Are you a veteran? </LABEL>
    <INPUT type="checkbox" id="veteran"  name="isVeteran">
    <BR>
    <BR>
    <INPUT type="submit" value="REGISTER NEW USER">
</FORM>
```

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Form Action & Method

```
<FORM action="/register.php">
```

```
<FORM action="#" method="get">
```

Methods

GET method:

- The data is appended to the URL request, after the ? (https://www.amazon.com/s?k=lenovo+laptops)
- Limitations: only send text data.

POST method:

- useful if you are sending sensitive information
- Upload binary information to the server (for example, images, pdfs, etc.)

Client-Side VS. Server-Side Programming Languages

Client-Side: HTML, CSS, JavaScript, and Bootstrap

Executed at the client (browser) side

Server-Side: C#, ASP.Net Core

Client send request, Executed at the server side, server send response to client

The <object> Element

The <object> element in HTML is used to embed external resources such as images, videos, audio, PDFs, or even other webpages into a document. It can also be used to embed multimedia content or interact with plugins (such as Flash or Java applets, though these are less common now).

```
<object data="file-url" type="media-type">
  Fallback content for browsers that do not support the object.
</object>
```

Example 1: Embedding an Image

- <object data="image.jpg" type="image/jpeg">
- Your browser does not support embedding images.
- </object>

Example 2: Embedding a PDF

- <object data="document.pdf" type="application/pdf" width="600" height="400">
- Your browser does not support PDFs. <a href="document.pdf">Download the PDF</a>.
- </object>

Bookmarks Embassy of Japan in... Inbox (234) - alihuss... Google

Practical Labs for

BIS401

WEBSITES DESIGN

For Business Information Systems Students

Example of Embedding a Video with <object>:

```
<object data="video.mp4" type="video/mp4" width="640" height="360"> <p>Your browser does not support embedded videos. You can <a href="video.mp4">download the video</a> instead.</p> </object>
```

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Lecture 1: Summary

HTML define the syntax and placement of special, embedded directions that aren't displayed by the browser but advise it how to display the contents of the document, including text, images, and other support media.

HTML Tags: text, Multimedia, …

Client-Side VS. Server-Side Programming Languages